

Université Côte d'Azur

Master 1 in Bioinformatics and Computational Biology (BBC)

Internship Report

Identification and Bioinformatic Analysis of Ion Channels That May Modulate Immunity and/or Extracellular Matrix in PDAC Stroma

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) is a highly desmoplastic cancer in which tumor cells coexist with a dense and heterogeneous stromal compartment. Because fibroblast states are linked to extracellular matrix remodeling, tissue architecture, and immune signaling, ion channels expressed in this compartment may contribute to biologically relevant programs beyond tumor cells alone. In this report, a multi-level bioinformatic workflow was used to identify and interpret potassium channel candidates associated with PDAC stroma. The first analytical block relied on a stromal single-cell RNA sequencing (scRNA-seq) dataset to prioritize KCN genes across complementary cell-level and pseudobulk differential expression strategies. This step highlighted *KCNMA1* as a candidate associated with an activated contractile stromal pole and *KCNJ8* as a candidate associated with a more quiescent or resident-like pole. No equally robust leader emerged for the inflammatory cancer-associated fibroblast (iCAF) axis. A second validation step used a broader PDAC single-nucleus RNA sequencing (snRNA-seq) atlas, referred to here as the USER dataset, to reinsert these candidates into the full tumor microenvironment. This step showed that *KCNN4* behaves as a strong tumor epithelial counterpoint, whereas *KCNMA1* and *KCNJ8* remain fibroblast-oriented. A local bulk RNA-seq model of BKC_a silencing in cancer-associated fibroblasts (CAFs) showed that *KCNMA1* knockdown modifies the transcriptome and is associated with lower myCAF- and iCAF-related scores. Finally, spatial transcriptomic analyses placed *KCNMA1*, *KCNJ8*, and *KCNN4* back into tissue context and supported a stronger association of *KCNMA1* with CAF-related territories and of *KCNN4* with tumor epithelial territories. Overall, this report proposes a simple stromal-centered model structured around *KCNMA1*, *KCNJ8*, and *KCNN4*.

Contents

Acknowledgements	i
Abstract	ii
List of Abbreviations and Definitions	v
1 Introduction	1
2 Objectives	2
3 Materials and Methods	2
3.1 Overall Bioinformatic Workflow	2
3.2 Moffitt Stromal scRNA-seq Discovery Block	3
3.3 USER snRNA-seq Validation Block	3
3.4 Bulk RNA-seq Analysis of BKC _a Silencing in CAFs	4
3.5 Spatial Transcriptomics and Signature Projection	4
3.6 Statistics and Multiple Testing	5
4 Results and Discussion	6
4.1 A Stromal KCN Core Was Identified in the Moffitt Dataset	6
4.2 USER Validation Repositioned the Candidates Along a Tumor–Stroma Axis	8
4.3 BKC _a Knockdown in CAFs Altered Programs Linked to Activated Stroma	10
4.4 Spatial Analyses Distinguished Stromal and Epithelial Territories	11
4.5 Signature Projection Strengthened the Contrast Between KCNMA1 and KCNN4	13
4.6 Cross-Dataset Limits and Interpretation of Low Ion Channel Expression	14
5 Conclusions and Perspectives	15
Clinical Opening and Limitations	16
References	17
M1 Skills Summary	18

List of Abbreviations and Definitions

BH Benjamini-Hochberg procedure.

BKCa Large-conductance calcium-activated potassium channel complex.

CAF Cancer-associated fibroblast.

csCAF Complement-secreting cancer-associated fibroblast.

DEG Differentially expressed gene.

ECM Extracellular matrix.

EMT Epithelial-mesenchymal transition.

FDR False discovery rate.

FFPE Formalin-fixed paraffin-embedded.

GSEA Gene set enrichment analysis.

iCAF Inflammatory cancer-associated fibroblast.

ifCAF Interferon-associated cancer-associated fibroblast.

KCN Potassium channel gene family.

log2FC Log₂ fold change.

myCAF Myofibroblastic cancer-associated fibroblast.

PCA Principal component analysis.

PDAC Pancreatic ductal adenocarcinoma.

PSC Pancreatic stellate cell.

Pseudobulk Aggregation of single-cell or single-nucleus counts by biological sample to perform sample-level statistics.

qPSC Quiescent pancreatic stellate cell.

RNA-seq RNA sequencing.

scRNA-seq Single-cell RNA sequencing.

shBKCa Short-hairpin RNA-mediated BKCa silencing.

smPSC Smooth muscle-like pancreatic stellate cell.

snRNA-seq Single-nucleus RNA sequencing.

ssGSEA Single-sample gene set enrichment analysis.

UMAP Uniform Manifold Approximation and Projection.

USER Project name used here for the Peng et al. PDAC snRNA-seq validation atlas.

1. Introduction

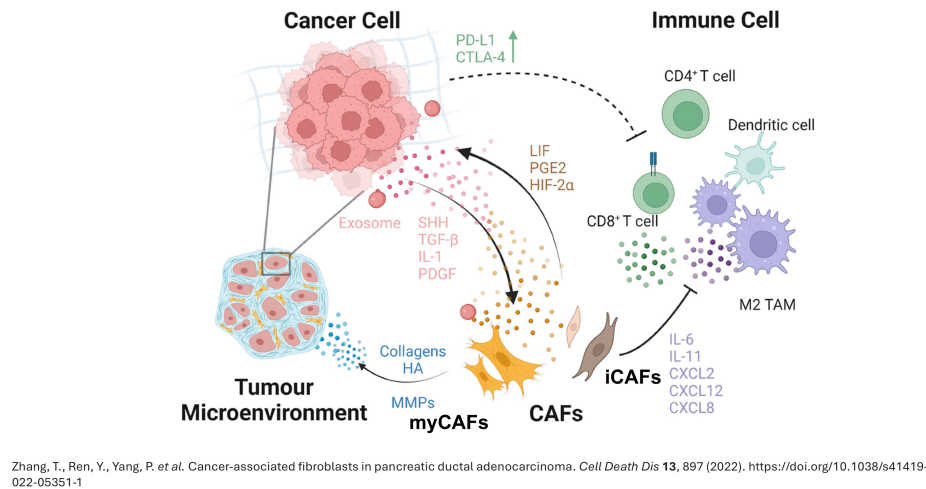


Figure 1: Fig. 1 | Biological context of the PDAC stromal microenvironment. PDAC epithelial cells are embedded in a dense stromal microenvironment structured by fibroblast heterogeneity, extracellular matrix remodeling, and immune interactions. The scheme introduces the myCAF/iCAF framework and the activated-versus-resident stromal polarity used throughout the report.

Pancreatic ductal adenocarcinoma (PDAC) is not only a disease of malignant epithelial cells. It is also a disease of tissue organization. One of its main features is a strong desmoplastic reaction in which tumor cells coexist with abundant extracellular matrix (ECM), immune cells, and heterogeneous fibroblasts. This stromal compartment affects tumor growth, limits drug penetration, and shapes local immune regulation (Ohlund et al., 2017; Elyada et al., 2019; Oh et al., 2023). For this reason, understanding stromal biology is important for understanding PDAC itself.

Cancer-associated fibroblasts (CAFs) are now recognized as a heterogeneous population rather than a single cell type. In PDAC, myofibroblastic CAFs (myCAFs) are usually described as matrix-rich and contractile, whereas inflammatory CAFs (iCAFs) are more secretory (Ohlund et al., 2017; Elyada et al., 2019). Recent single-cell studies also support the existence of more quiescent or resident-like stellate states and additional inflammatory or interferon-associated fibroblast programs (Hosein et al., 2019; Oh et al., 2023). In this report, this biological complexity is simplified into a practical contrast between an activated contractile stromal pole and a more quiescent resident-like stromal pole.

Ion channels are relevant in this context because they regulate membrane potential, calcium entry, secretion, cell volume, and mechanical responses. In fibroblasts, these functions can influence contraction, matrix deposition, and communication with neighboring cells. However, ion channels are often expressed at relatively low transcript levels, especially in sparse single-cell or spatial assays, which makes them easy to miss if analysis relies only on absolute expression. This does not mean that they are biologically unimportant. A limited number of ion channels can still produce strong electrophysiological effects on membrane potential and calcium signaling, and

these effects can translate into major functional changes (Scruggs et al., 2020; Jiang et al., 2019). In this project, I focused on the potassium channel family because it is the largest ion channel family and because potassium channel biology is a clear area of interest in the host laboratory.

The broader project initially identified a union of 25 KCN candidates across the stromal discovery analyses. Because this report is limited in size and number of figures, I focus on three genes that best structure the story. *KCNMA1*, which encodes the pore-forming subunit of the BKCa channel, is used as the main activated stromal candidate. *KCNJ8* is used as a candidate associated with a more quiescent or resident stromal pole. *KCNN4* is used as a tumor epithelial counterpoint that helps distinguish stromal and epithelial territories. Other KCN genes remain interesting and will be discussed briefly in the final schematic and perspectives.

2. Objectives

The main objective of this internship was to identify ion channels that may modulate extracellular matrix organization and/or immune-related stromal functions in PDAC. To address this question, the report followed four linked goals. First, I aimed to prioritize stromal potassium channel candidates in a stromal scRNA-seq dataset. Second, I aimed to validate and reposition these candidates in a broader PDAC snRNA-seq atlas containing both stromal and tumor compartments. Third, I aimed to test whether BKCa silencing in CAFs is associated with coherent transcriptomic changes. Fourth, I aimed to place the main candidates back into tissue space using spot-based spatial transcriptomics. The final goal was not to claim a complete mechanistic model, but to build a coherent and readable stromal-centered interpretation supported by several data layers.

3. Materials and Methods

3.1. Overall Bioinformatic Workflow

This work is presented as a workflow rather than as a finalized pipeline. It combines four analytical blocks that match the main figures of the report: stromal single-cell prioritization, external validation in a broader snRNA-seq atlas, local bulk RNA-seq perturbation around BKCa signaling, and spatial transcriptomic localization. The workflow is currently being reorganized into a more reusable pipeline for later use in the project.

The project relied mainly on **R** and **Python**, with command-line orchestration through **Bash**- or shell-based scripts. The main R packages were **Seurat** (single-cell analysis framework), **DESeq2** (count-based differential expression for RNA-seq), **MAST** (single-cell hurdle-model differential expression), **fgsea** (fast gene set enrichment analysis), **msigdbR**, **escape**, **ggplot2**, **ggrepel**, **VennDiagram**, **pheatmap**, and standard data-handling packages such as **dplyr** and **tibble**. The main Python packages were **scanpy** (single-cell analysis toolkit), **pandas**, and the **Hotspot** framework. All versioned scripts, figure-generation workflows, and extended project details are organized as a reproducible analysis structure available in the accompanying GitHub repository.

3.2. Moffitt Stromal scRNA-seq Discovery Block

The first analytical block used a stromal-focused Seurat object derived from the coordinated single-cell PDAC resource from Oh et al. (Oh et al., 2023). Single-cell RNA sequencing (scRNA-seq) measures the transcriptome of individual whole cells after dissociation. The full atlas contains several compartments, including tumor, immune, endothelial, and stromal populations, but the report starts from the stromal object because the internship question is stromal-centered. A broader tumor-versus-stroma comparison was addressed later with the USER atlas, which is more suitable for that purpose.

The stromal object contained 21,469 stromal cells from 40 samples and 29 patients, distributed across eight stromal cell types. UMAP plots were read from the Seurat object and are based on normalized single-cell expression values. The object also contained the original activated-stroma and normal-stroma scores from the Moffitt framework. I used the difference between these two scores to define a broad **StromaState** variable. This was useful because it allowed the Venn diagram to include both the original activated-versus-normal stromal view and finer subtype-specific contrasts.

Four complementary differential expression strategies were combined in Fig. 2b. Test 1 was a patient-aware pseudobulk **DESeq2** comparison between activated-like and normal-like stromal states. In this test, raw counts were aggregated by patient and state before running a count-based negative binomial model with a Wald test (Love et al., 2014). Test 2 was a patient-aware pseudobulk **DESeq2** subtype-versus-rest comparison, again using aggregated raw counts. Test 3 was a cell-level hurdle-style model designed to separate gene detection from positive-expression behavior. Such models are useful in zero-inflated data because a gene can differ either by being detected more often or by being more strongly expressed when detected (Finak et al., 2015). Test 4 was a standard **MAST** analysis using log-normalized single-cell expression values (Finak et al., 2015). The Venn diagram therefore summarizes convergence across different input matrices and different statistical assumptions.

For the direct qPSC-versus-myCAF comparison shown in Fig. 2d, raw counts were aggregated into patient-aware pseudobulk samples. Only paired patients with enough cells in both subtypes were retained, leading to 29 paired patients, 5,386 qPSC cells, and 4,699 myCAF cells. **DESeq2** was run with the design `~ patient + subtype`. Hallmark gene set enrichment analysis (GSEA) was then performed with **fgsea** using the Wald statistic ranking (Love et al., 2014; Korotkevich et al., 2021).

3.3. USER snRNA-seq Validation Block

The second analytical block used a broader PDAC single-nucleus RNA sequencing (snRNA-seq) dataset from Peng et al. (Peng et al., 2020). snRNA-seq measures nuclear RNA instead of whole-cell RNA and is often more stable in fibrotic tissues. In this report, this dataset is called the USER dataset, following the project naming convention. The key advantage of this dataset is that it contains the full microenvironment rather than fibroblasts alone, which makes it useful

for placing stromal candidates back into a tumor-versus-stroma context.

The global USER figure was built from published metadata and UMAP coordinates. The dataset contained 88,031 nuclei, including 17,537 fibroblast nuclei. Patient composition was summarized directly from metadata because strong inter-patient variability was expected and needed to be made explicit before any tumor-versus-fibroblast comparison, as later shown in Fig. 3b.

For the tumor-versus-fibroblast contrast, raw counts were aggregated by patient and cell subset. Only patients containing both tumor and fibroblast nuclei were kept. Genes with very low total counts were removed before analysis. DESeq2 was then run with the design `~ patient + group`. The final contrast included 15 paired patients, 43,817 tumor nuclei, 17,537 fibroblast nuclei, and 18,993 tested genes.

3.4. Bulk RNA-seq Analysis of BKCa Silencing in CAFs

The third analytical block used a local bulk RNA-seq dataset generated in the host laboratory. In this experiment, BKCa was silenced by short-hairpin RNA in cultured CAFs, creating a **shBKCa** condition compared with non-targeting controls. In this report, BKCa refers to the large-conductance calcium-activated potassium channel complex centered on the pore-forming *KCNMA1* gene and interpreted together with the associated beta-subunit genes *KCNMB1* and *KCNMB4*. Bulk RNA-seq measures the average transcriptome of each sample rather than individual cells. The comparison included four control samples and four **shBKCa** samples, and gene-level counts were collapsed into 17,889 genes.

Principal component analysis (PCA) was used as an internal quality-control step on normalized transformed expression values to confirm separation of replicates by condition, although this PCA is not shown in the main report figures. Differential expression was performed on raw count data with DESeq2 (Love et al., 2014). Hallmark pathway enrichment was performed with fgsea (Korotkevich et al., 2021). CAF signature heatmaps were built from normalized expression values scaled by gene across samples, which is why the legend corresponds to a row z-score. At signature level, condition differences were summarized with Wilcoxon rank-sum tests. The myCAF and iCAF signatures were based on published PDAC CAF states, mainly from Ohlund et al. and Elyada et al. (Ohlund et al., 2017; Elyada et al., 2019).

3.5. Spatial Transcriptomics and Signature Projection

The spatial block used a Visium FFPE PDAC dataset from Khaliq et al. (Khaliq et al., 2024). Spatial transcriptomics measures RNA expression at spot level while preserving tissue coordinates. In this dataset, the 30 tissue sections represented 91,496 spots in total, including 3 normal pancreas sections (9,820 spots), 10 PDAC sections (35,458 spots), 12 liver metastasis sections (28,520 spots), and 5 lymph node sections (17,698 spots). Structural annotations and multicellular ecotypes from the reference study were reused as a tissue framework.

In this report, I focused mainly on normal pancreas and primary PDAC sections. The reason is simple: the internship question concerns pancreatic stromal organization, and the figure budget

is limited. Liver metastasis and lymph node sections remain useful in the larger project, but they are not necessary to support the main stromal-centered argument developed here.

Spatial interpretation relied on two complementary levels. First, local Hotspot-style maps were used to visualize spatially coherent territories for *KCNMA1* and *KCNN4*. In simple terms, the **Hotspot** framework detects locally structured patterns rather than isolated high spots, which makes it useful for distinguishing real tissue territories from scattered signal (DeTomaso and Yosef, 2021). Second, reference signatures were projected at spot level and compared with KCN signals using Spearman rank correlation. Spearman correlation was preferred because spot-level values are sparse, non-Gaussian, and better interpreted through monotonic association than strict linearity.

The tumor epithelial signature used in Fig. 6a was defined in the laboratory from epithelial markers considered informative in this project: *EPCAM*, *MUC1*, *KRT7*, and *KRT8*, with *KRT18* retained when available in the assay. The CAF signature used in Fig. 6b came from the reference spatial study and was projected as a paper-derived ssGSEA-like score (Khaliq et al., 2024). In practice, the spatial analysis combined R- and Python-based processing using **Seurat**, **escape**, **scanpy**, **pandas**, and **Hotspot**.

3.6. Statistics and Multiple Testing

Effect size is mainly described in this report with log2FC, and statistical significance is summarized with adjusted p-values or false discovery rate (FDR). In **DESeq2**, p-values came from Wald tests on count-based negative binomial models (Love et al., 2014). In **MAST**, p-values came from a hurdle framework applied to log-normalized expression (Finak et al., 2015). Because omics analyses test many genes and pathways at the same time, multiple testing correction was performed with the Benjamini-Hochberg (BH) method rather than Bonferroni correction. BH was preferred because it controls the expected proportion of false discoveries while preserving reasonable sensitivity in discovery-oriented transcriptomic analyses.

Single-cell and single-nucleus differential results were interpreted together with pseudobulk analyses to avoid over-reliance on cell-level pseudoreplication. Wilcoxon tests were used for non-parametric comparison of summary scores, such as CAF signatures in bulk RNA-seq. Spearman correlation was used in the spatial block for spot-level association analyses.

4. Results and Discussion

4.1. A Stromal KCN Core Was Identified in the Moffitt Dataset

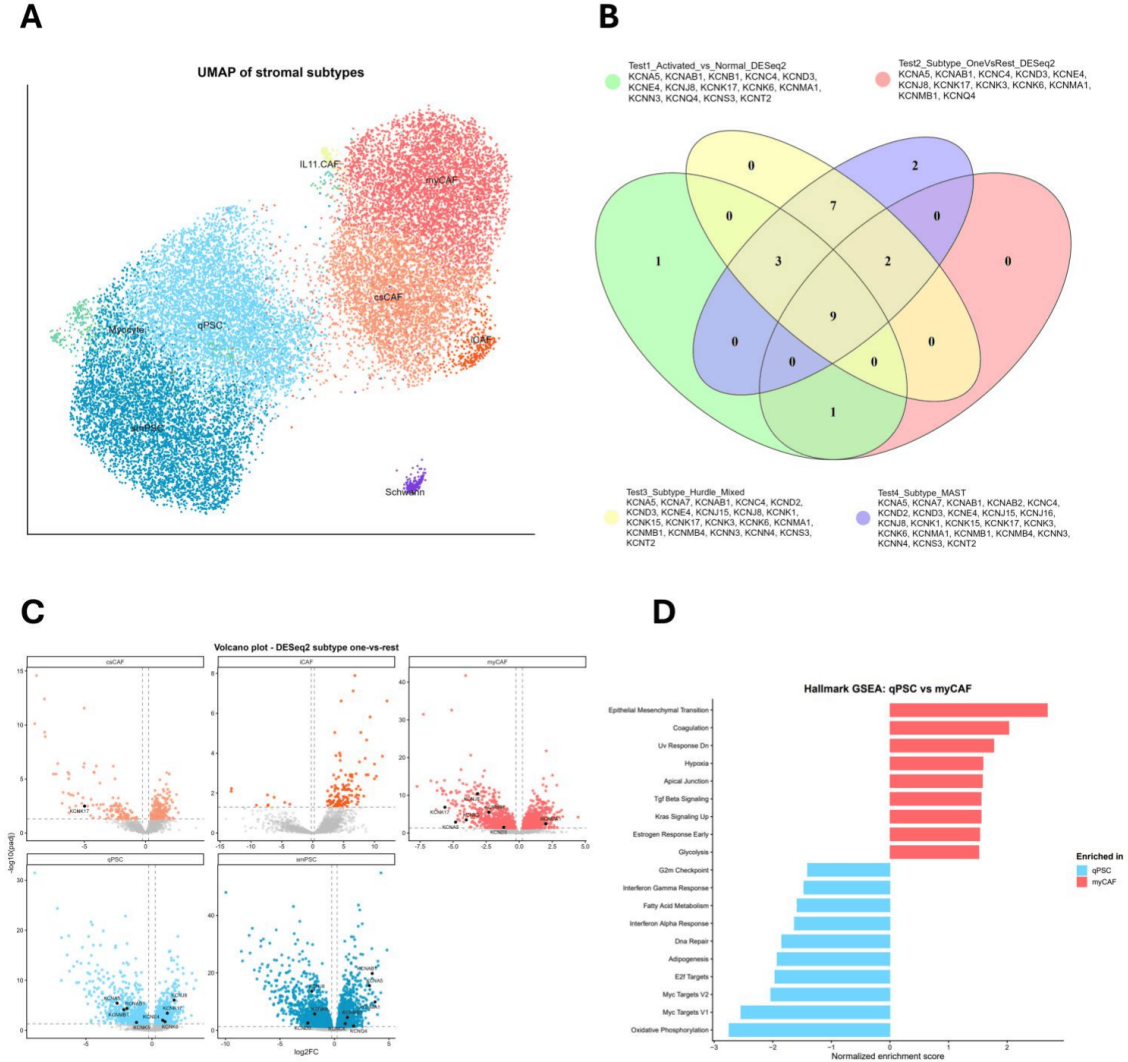


Figure 2: Fig. 2 | Stromal single-cell prioritization in the Moffitt dataset. **a** UMAP representation of stromal cell subtypes used as the reference map for the stromal part of the report. **b** Four-test Venn summary showing overlap between complementary differential expression strategies used to prioritize KCN candidates across cell-level and pseudobulk approaches. **c** Volcano plots from the pseudobulk subtype-versus-rest analysis highlighting KCN behavior across stromal subtypes. **d** Hallmark GSEA comparison between qPSC and myCAF.

In this section, as shown in Fig. 2a, the stromal compartment is not a homogeneous fibroblast cloud. It is organized into qPSC, smPSC, myCAF, csCAF, and iCAF states, together with smaller associated populations. This matters because the project question is not only whether a KCN gene is stromal, but also which stromal state it follows. The stromal object contained 21,469 cells in total, including 5,743 qPSCs, 5,521 smPSCs, 4,809 myCAFs, 4,242 csCAFs, and only 338 iCAFs.

As shown in Fig. 2b, the four differential expression strategies converge on a compact robust

core. The full union of candidates was useful to preserve potentially relevant KCN genes at an exploratory stage, but the strict intersection is the most robust part of the result. In practice, nine KCN genes were detected by all four methods. This convergence is important because the four methods do not test exactly the same thing. The two pseudobulk tests are conservative and sample-aware. The hurdle-based single-cell tests are more sensitive to sparse detection behavior. The **StromaState** comparison was also worth keeping because it reconnects the subtype-based analysis to the original activated-versus-normal stromal logic from the Moffitt framework.

As shown in Fig. 2c, the subtype-specific effect sizes refine this convergence. In the pseudobulk subtype-versus-rest analysis, *KCNMA1* was strongly enriched in smPSC with $\log_2\text{FC} = 3.73$ ($\text{FDR} = 1.30 \times 10^{-10}$) and was also enriched in myCAF with $\log_2\text{FC} = 2.00$ ($\text{FDR} = 3.63 \times 10^{-3}$). In contrast, *KCNJ8* was enriched in qPSC with $\log_2\text{FC} = 1.67$ ($\text{FDR} = 9.11 \times 10^{-7}$) and strongly depleted in myCAF with $\log_2\text{FC} = -3.16$ ($\text{FDR} = 4.04 \times 10^{-11}$). *KCNN4* did not emerge as a significant stromal subtype marker in this stromal-restricted setting. These values align with the primary stromal polarity established in this report, wherein *KCNMA1* delineates an activated, contractile pole, whereas *KCNJ8* characterizes a more quiescent, resident-like phenotype.

As shown in Fig. 2d, this polarity is not limited to single genes. The myCAF side is enriched in epithelial-mesenchymal transition, TGF- β signaling, coagulation, hypoxia, glycolysis, and KRAS signaling up. In contrast, qPSC is enriched in oxidative phosphorylation, fatty acid metabolism, interferon response, and DNA repair. The iCAF axis was explored, but it remained less robust in this dataset. This is consistent with the low number of iCAF cells. For example, *KCNJ8* showed $\log_2\text{FC} = -3.34$ in iCAF but remained non-significant after correction ($\text{FDR} = 0.311$), and *KCNMA1* showed a modest positive signal without significance. In this report, iCAF is therefore treated as an explored but unresolved axis rather than as a fourth strong KCN-centered pole.

4.2. USER Validation Repositioned the Candidates Along a Tumor–Stroma Axis

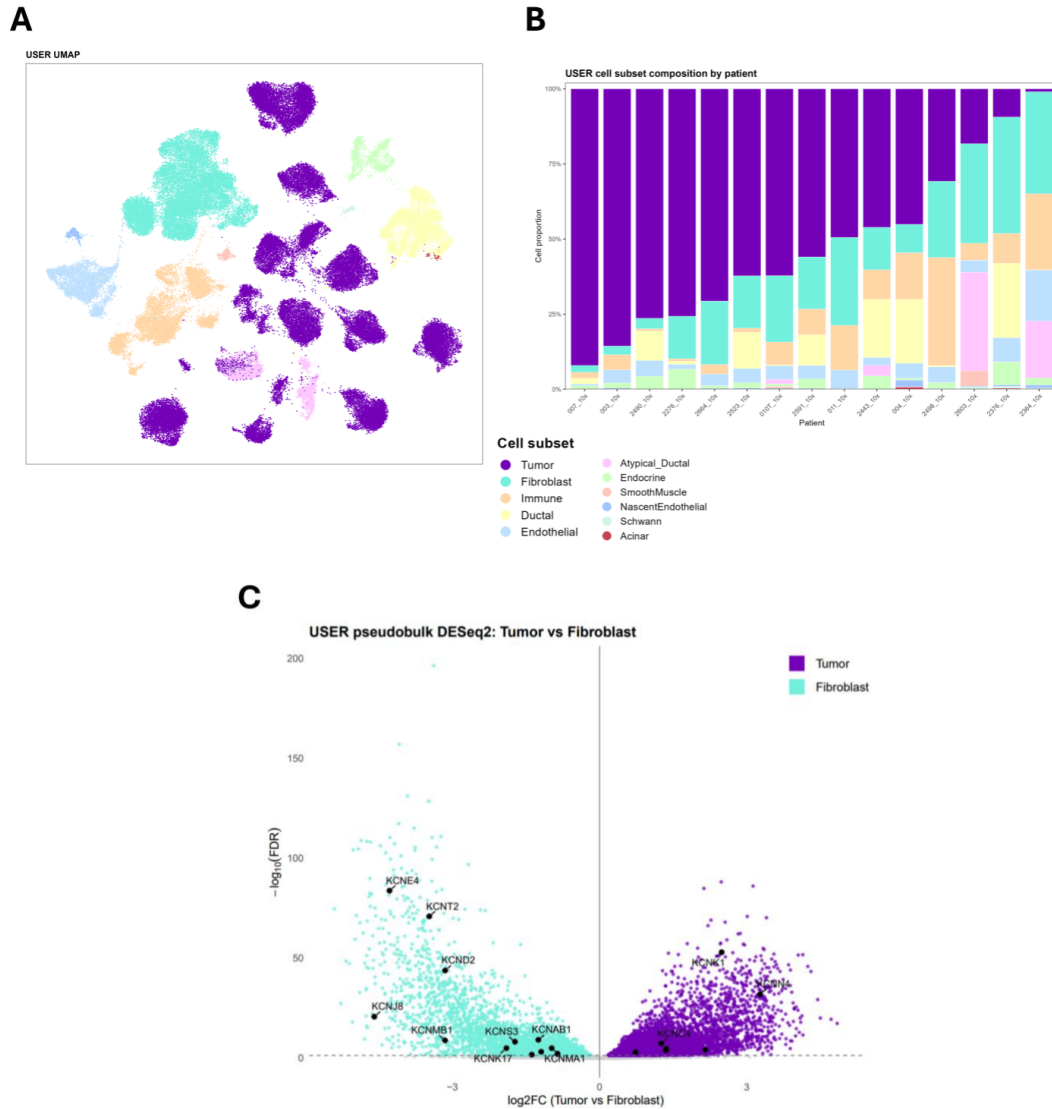


Figure 3: Fig. 3 | External validation in the USER snRNA-seq dataset. a Global UMAP of the USER PDAC atlas used to reinsert stromal candidates into a broader tumor microenvironment context. **b** Cell-type composition across patients, shown to highlight inter-patient variability and to justify patient-aware pseudobulk analyses. **c** Volcano plot of the pseudobulk Tumor-versus-Fibroblast comparison, used to position the selected KCN candidates along a stromal-versus-tumor axis.

In this section, as shown in Fig. 3a, the candidates are repositioned in a broader whole-tissue context. This step is important because a gene that looks stromal inside a fibroblast-focused object can still be shared with other compartments when the whole tumor microenvironment is considered. The USER atlas is particularly useful for this because it contains tumor, fibroblast, immune, endothelial, and ductal compartments in the same framework.

As shown in Fig. 3b, there is strong inter-patient variation in cellular composition. Some samples are largely tumor-dominated, whereas others contain larger fibroblast fractions. This directly justifies the patient-aware pseudobulk design used for the tumor-versus-fibroblast contrast. Without this design, large patient-specific composition effects could be mistaken for true gene-specific

biology.

As shown in Fig. 3c, the simplest global question is whether a gene is closer to tumor or fibroblast in the full atlas. In this direct Tumor-versus-Fibroblast comparison, *KCNN4* was strongly tumor-up, with $\log_2\text{FC} = 3.27$ and $\text{FDR} = 1.23 \times 10^{-32}$. In contrast, *KCNJ8* was strongly fibroblast-up, with $\log_2\text{FC} = -4.59$ and $\text{FDR} = 1.92 \times 10^{-21}$. *KCNMA1* also remained on the fibroblast side, with $\log_2\text{FC} = -0.975$ and $\text{FDR} = 1.13 \times 10^{-5}$, although the effect size was smaller than for *KCNJ8*. This result is very informative for the general model. It confirms *KCNN4* as a tumor epithelial counterpoint, keeps *KCNJ8* as a strongly fibroblast-oriented gene, and places *KCNMA1* in a stromal-centered but more microenvironmental position.

4.3. BKCα Knockdown in CAFs Altered Programs Linked to Activated Stroma

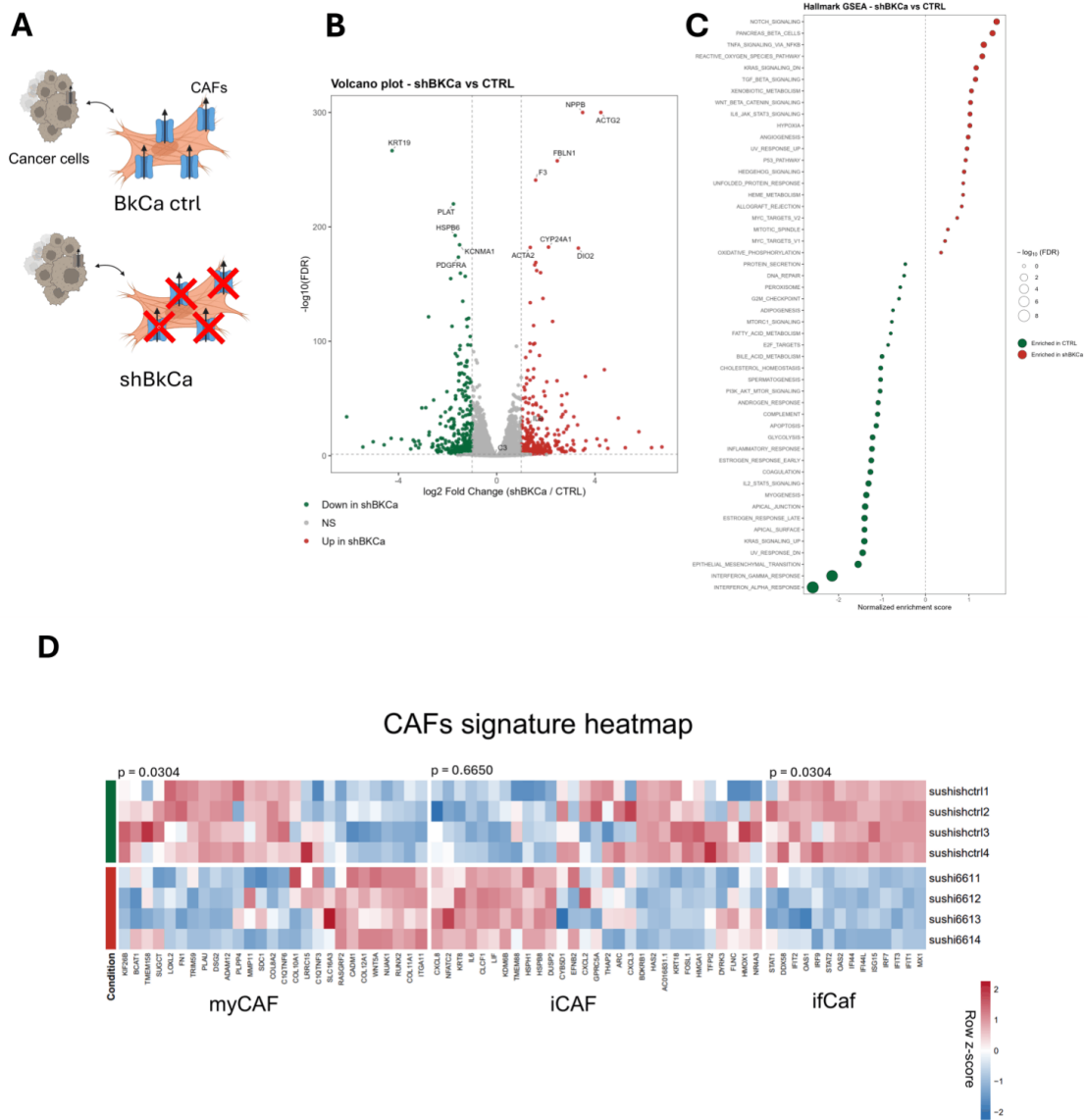


Figure 4: Fig. 4 | Bulk RNA-seq perturbation analysis in a BKCα-centered CAF model. a Experimental overview of the control and shBKCα comparison used in the local bulk RNA-seq block. **b** Volcano plot of differential expression between shBKCα and control samples. **c** Hallmark GSEA summary used to identify broad transcriptomic programs associated with the perturbation. **d** Heatmap of selected CAF-related signatures, including myCAF-, iCAF-, and ifCAF-related markers, used to examine whether the perturbation is associated with stromal program shifts.

In this section, as shown in Fig. 4a, the question is no longer where a gene is expressed, but whether BKCα silencing in CAFs changes the transcriptome in a coherent way. This part is important because it adds a functional layer from the host laboratory to the descriptive multi-omics analyses.

As shown in Fig. 4b, the perturbation worked. *KCNMA1* itself was reduced in shBKCα, with $\log_2\text{FC} = -1.51$ and $\text{FDR} = 4.45 \times 10^{-185}$. The volcano plot shows changes in structural or stromal genes such as *ACTA2*, *ACTG2*, *FBLN1*, and *PDGFRA*, supporting the idea that BKCα

signaling is linked to stromal cell state.

At pathway level, as shown in Fig. 4c, the strongest significant enrichments are found on the control side, especially interferon alpha response, interferon gamma response, and epithelial-mesenchymal transition. In other words, the major effect of BKC_a silencing in this model is not the appearance of one dominant new program, but rather the weakening of already structured programs.

As shown in Fig. 4d, the CAF-oriented heatmap is especially useful for the stromal-centered interpretation because it suggests a global shift in CAF state rather than a single binary marker change. The clearest pattern is seen for the ifCAF-like signature, which is markedly lower in **shBKC_a** than in controls (mean score 0.84 in CTRL versus -0.84 in **shBKC_a**; nominal Wilcoxon $p = 0.0304$, BH-adjusted FDR = 0.076). The myCAF signature also tends to decrease in **shBKC_a** (mean score 0.12 in CTRL versus -0.12 in **shBKC_a**; $p = 0.0304$), but the heatmap indicates more heterogeneity across samples than for ifCAF. In contrast, the iCAF signature is more variable and does not show a robust directional shift ($p = 0.665$). Although limited by the small cohort size ($n = 8$), this pattern supports the idea that BKC_a is not only associated with one CAF marker set, but more broadly acts as a driver of CAF signatures and CAF plasticity.

4.4. Spatial Analyses Distinguished Stromal and Epithelial Territories

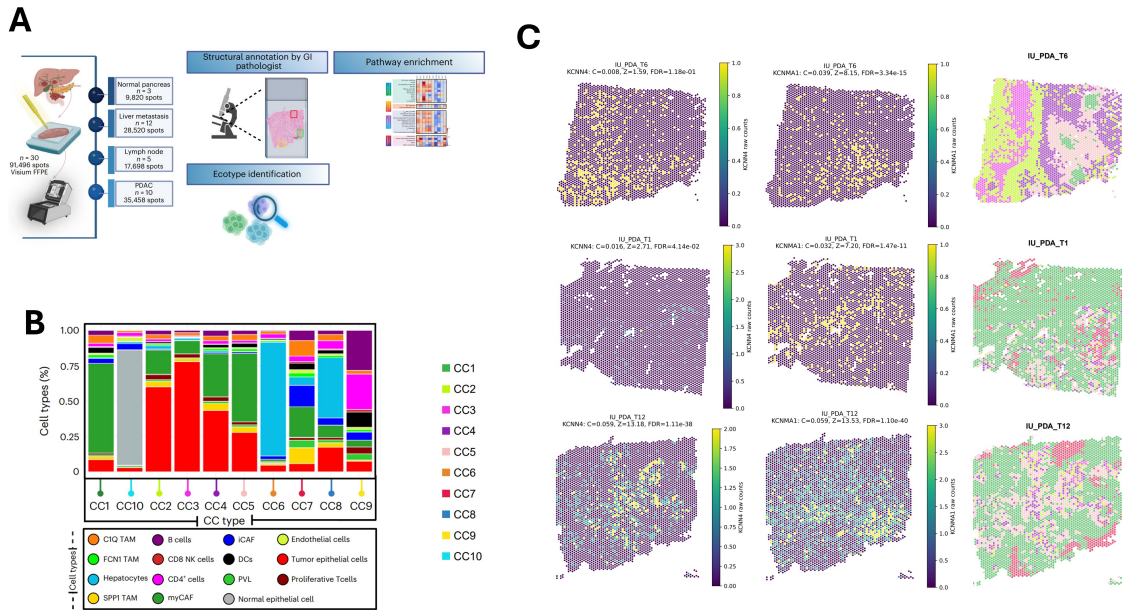


Figure 5: Fig. 5 | Spatial transcriptomic context for the main KCN candidates. **a** Overview of the spatial transcriptomic dataset, including section types, spot numbers, and ecotype-related information used in this part of the report. **b** Ecotype composition summary from the reference PDAC spatial study. **c** Representative Hotspot maps for *KCNMA1* and *KCNN4* in selected PDAC sections (T6, T1, and T12), shown together with the neighboring ecotype context.

In this section, as shown in Fig. 5a, the analysis returns to tissue architecture. For the Hotspot results discussed in Fig. 5, the Z score corresponds to the standardized strength of spatial auto-

correlation for a gene within one section, C is the associated scaled autocorrelation coefficient, and FDR is the Benjamini-Hochberg-adjusted significance of this spatial test. Higher positive Z and C values therefore indicate a more coherent patch-like spatial pattern rather than diffuse expression. For the signature projections discussed in Fig. 6, ρ denotes the Spearman rank correlation between a KCN gene and the projected signature across spots in one section, and FDR again denotes the Benjamini-Hochberg-adjusted significance of that correlation (DeTomaso and Yosef, 2021). The spatial dataset, as shown in Fig. 5a spans several anatomical contexts, but as said before, the report focuses mainly on normal pancreas and primary PDAC sections because these are relevant here. Fig. 5b recalls the multicellular ecotypes described in the reference spatial study.

As shown in Fig. 5c, the main spatial contrast of this block is that *KCNMA1* appears broader and more reproducible than *KCNN4*. In T6, T1, and T12, *KCNMA1* is repeatedly detected in regions compatible with stromal or mixed microenvironmental territories (ecotypes CC1 and CC5 in the paper), whereas *KCNN4* is more section-dependent and visually closer to epithelial tumor territories (ecotypes CC2 and CC3). The ecotype maps next to these panels help interpret these signals as territory associations rather than pure single-cell identities. This distinction is important because Visium spots average transcripts from small multicellular neighborhoods.

This is also why *KCNJ8* remains in the final model even though Fig. 5 mainly emphasizes *KCNMA1* and *KCNN4*. Although it is not shown directly here, *KCNJ8* displayed a stronger spatial Hotspot signal in normal pancreas than in primary PDAC, with a median Z score of 5.17 across normal sections versus 1.15 across PDAC sections, together with a higher median C value (0.0238 versus 0.00617). In the report, *KCNJ8* is therefore mainly supported by the stromal polarity seen in Moffitt and confirmed in USER, while its spatial pattern also suggests relative enrichment in non-diseased pancreatic tissue compared with PDAC.

4.5. Signature Projection Strengthened the Contrast Between *KCNMA1* and *KCNN4*

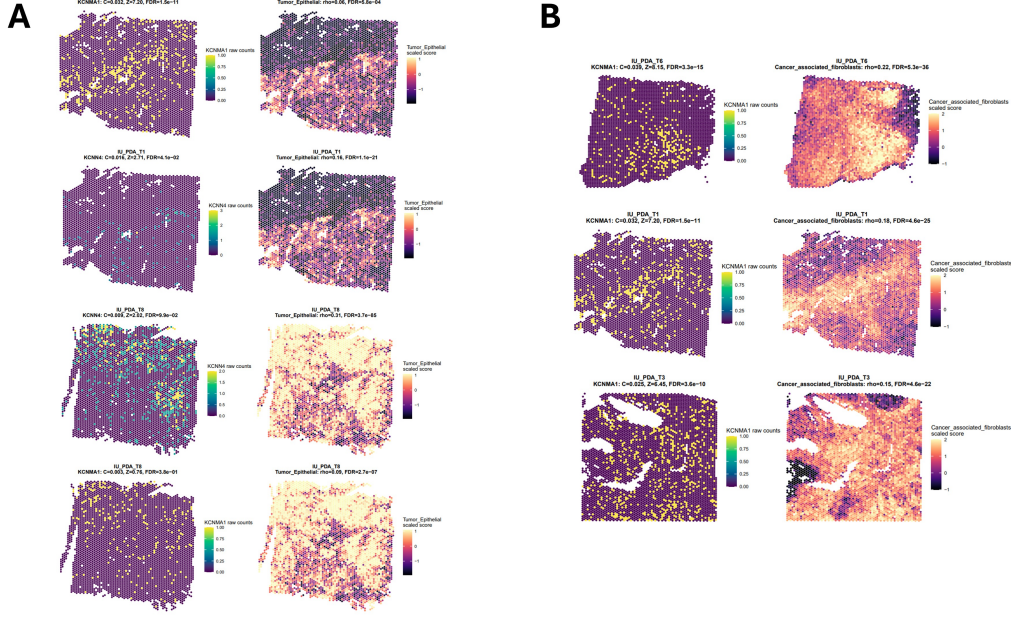


Figure 6: Fig. 6 | Spatial projection of tumor epithelial and CAF-related signatures. a Spot-level Spearman correlation projections comparing *KCNMA1* and *KCNN4* with a manually defined tumor epithelial signature in selected PDAC sections T1 and T8. **b** Spot-level comparison between *KCNMA1* and the cancer-associated fibroblast score in representative sections T6, T1, and T3.

In this section, as shown in Fig. 6a, the spatial reading is refined by comparing gene expression to continuous biological signatures. This approach is useful because a spot-level gene map alone can remain ambiguous, whereas a spot-level correlation with a known reference program gives a more quantitative interpretation.

As shown in Fig. 6a, *KCNN4* follows the manually defined tumor epithelial signature more clearly than *KCNMA1*. In T1, the tumor epithelial score correlated with *KCNN4* at $\rho = 0.16$ ($\text{FDR} = 1.1 \times 10^{-21}$), whereas the correlation with *KCNMA1* was lower at $\rho = 0.06$ ($\text{FDR} = 5.8 \times 10^{-4}$). In T8, the same trend remained visible, with $\rho = 0.31$ for *KCNN4* ($\text{FDR} = 3.7 \times 10^{-85}$) and $\rho = 0.09$ for *KCNMA1* ($\text{FDR} = 2.7 \times 10^{-7}$). These values are consistent with the USER tumor-versus-fibroblast result and strengthen the interpretation of *KCNN4* as a tumor epithelial counterpoint.

As shown in Fig. 6b, *KCNMA1* provides the converse information. Across the sections shown, *KCNMA1* positively tracked the CAF score, with $\rho = 0.22$ in T6 ($\text{FDR} = 5.3 \times 10^{-36}$), $\rho = 0.18$ in T1 ($\text{FDR} = 4.6 \times 10^{-25}$), and $\rho = 0.15$ in T3 ($\text{FDR} = 4.6 \times 10^{-22}$). This does not imply exclusive expression in every fibroblast-containing spot, but it strongly supports a preferential association with stromal and CAF-related territories.

Taken together, Figures 5 and 6 provide the final spatial answer of the report. *KCNN4* is better aligned with tumor epithelial territories, whereas *KCNMA1* is better aligned with stromal and CAF-rich territories. This spatial contrast completes the multi-level interpretation built from

Moffitt, USER, and the local bulk perturbation model.

4.6. Cross-Dataset Limits and Interpretation of Low Ion Channel Expression

An important limitation of this project is that ion channels are often lowly expressed at transcript level across all the data types used here. This is true in scRNA-seq, where dropout and sparse counts reduce detection sensitivity; in snRNA-seq, where nuclear capture can underrepresent some transcripts; in bulk RNA-seq, where stromal signals are averaged across the whole sample; and in spatial transcriptomics, where spot-level averaging dilutes rare or localized signals. For this reason, absolute expression alone is not a sufficient criterion to evaluate the relevance of ion channels in PDAC.

This limitation is especially important for potassium channels because their biological importance does not scale linearly with transcript abundance. Even a modest number of channels can shift membrane potential, calcium entry, cell volume regulation, or contractile behavior. In fibroblasts and myofibroblasts, these functions can directly affect matrix remodeling and tissue mechanics (Scruggs et al., 2020). In tumor cells, channels such as *KCNN4* can support calcium-dependent signaling and tumor progression (Jiang et al., 2019). In other words, low transcript abundance does not mean low functional significance.

This is why the workflow was intentionally built around convergence rather than single evidence. Instead of relying only on raw expression level, the project combined differential expression, patient-aware pseudobulk analysis, pathway association, perturbation response, and tissue localization. This does not remove all uncertainty, but it provides a more balanced way to study low-abundance but potentially high-impact genes.

5. Conclusions and Perspectives

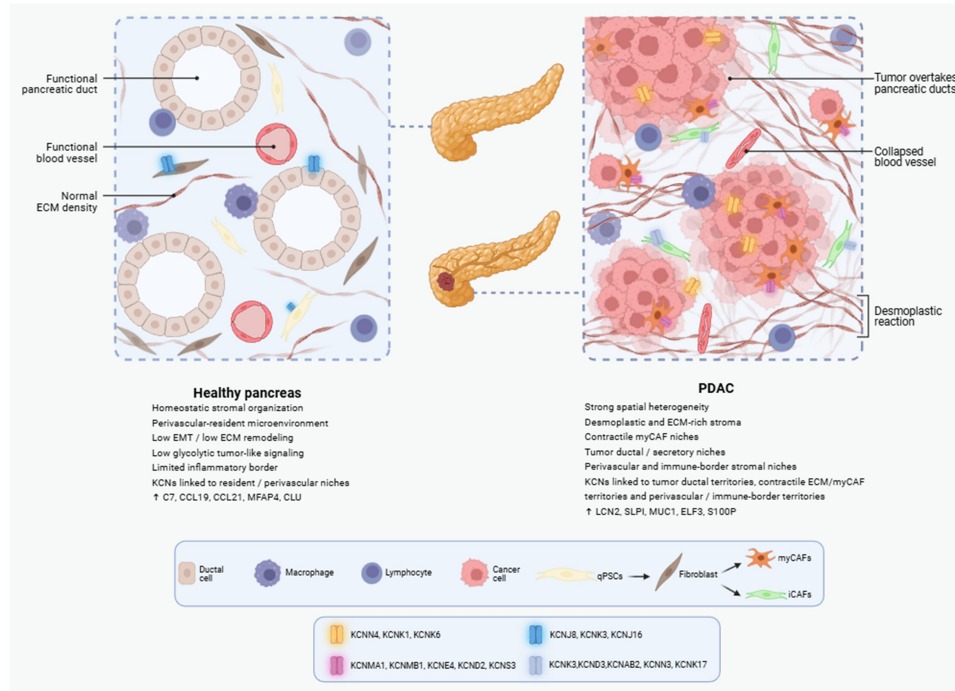


Figure 7: Fig. 7 | Final schematic summary of the selected KCN landscape in PDAC. The scheme summarizes the proposed relationship between PDAC stroma, tumor territories, and the main KCN candidates emphasized in this report, with a simplified comparison to non-diseased pancreas. The writing here focuses on *KCNMA1*, *KCNJ8*, and *KCNN4* because of report constraints, but the broader KCN union also suggests a directional organization of additional candidates across the PDAC microenvironment.

This report used a multi-level bioinformatic workflow to study potassium channel candidates in PDAC stroma. The main result is not the identification of one universal stromal channel, but the definition of a small and interpretable landscape. *KCNMA1* is the strongest candidate for an activated, matrix-rich, and contractile stromal pole. *KCNJ8* supports a more quiescent or resident-like stromal pole. *KCNN4* is the clearest tumor epithelial counterpoint and helps avoid an overly simplified all-stromal interpretation.

This model is coherent because it is supported by complementary data layers. The Moffitt stromal block provides the main stromal polarity. The USER atlas repositions the candidates in a broader tumor microenvironment. The local bulk RNA-seq model shows that BKCα silencing in CAFs is linked to measurable transcriptomic change. The spatial block shows that *KCNMA1* and *KCNN4* do not occupy the same biological territories in tissue.

The iCAF axis remains an important biological question, but it did not yield a KCN candidate as robust as *KCNMA1* or *KCNJ8* in this report. This should be interpreted as a limit of the current datasets and statistical power rather than as a proof that ion channels are irrelevant to iCAF biology.

Future work can extend this study in two directions. First, the current workflow can be formalized

(and will be) into a more reusable pipeline for additional ion channel families. Second, the strongest candidates can be tested experimentally in more direct stromal models to determine whether the *KCNMA1*/BKCa axis modulates fibroblast mechanics, matrix deposition, or stromal immune signaling. This perspective is especially interesting because the three-gene focus used here is a reporting choice, not a biological ceiling, and the broader KCN union already suggests a wider organization of potassium channel programs across the PDAC microenvironment.

Clinical Opening and Limitations

As a secondary extension, bulk survival analysis was performed to ask whether some prioritized KCN genes also show a clinical signal at patient level. In practice, this part combined a multicohort two-stage Cox meta-analysis and within-cohort Kaplan–Meier exploration. The results were interesting but remained modest. For example, higher *KCNT2* and *KCNMA1* tended to be associated with better outcome, whereas higher *KCNMB4* tended to be associated with worse outcome. However, no KCN remained significant after BH correction in the main meta-analysis, and the direction or strength of the signal could vary across cohorts. This part therefore supports the idea that some stromal or microenvironment-related KCN may leave a clinical trace, but it remains limited because bulk cohorts cannot resolve stromal localization, cannot separate stromal and tumor contributions directly, and cannot establish mechanism.

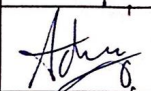
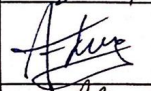
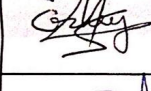
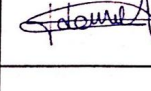
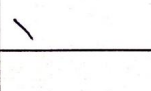
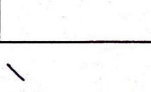
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M1 Skills Summary

During this internship, I learned how to organize a multi-level bioinformatic workflow, work with scRNA-seq, snRNA-seq, bulk RNA-seq, and spatial transcriptomic datasets, perform patient-aware pseudobulk analyses, and connect pathway enrichment results to an interpretable biological model. I also improved my ability to document analytical steps, generate readable figures, and discuss transcriptomic results with the level of caution required for low-abundance genes such as ion channels. More than just technical skills, this project showed me how exciting and complex cancer molecular biology is when studied through bioinformatics. I learned that data analysis is about asking the right statistical questions, searching for the proper tools, and staying critical of the results. It requires a lot of trial and error, trying again and again even when things do not work or look blurry at first, and remaining calm to find the answers (at least, try to get some). Ultimately, it taught me how to make important choices, justify them, and never stop questioning the data.

Seminar Attendance Sheet

Nom: <u>Mouroum</u>	Prénom: <u>Younna</u>	Master① ou 2: <u>B&C</u>
Nom du conférencier	Titre du séminaire	Nom et signature de l'hôte
Dr Yves Herveau Dr Gabriel Krouh Agathe Bayron Wendie Khouh 16/10/25	Conférence HODI	
Dr Danièle Rainaldi Dr Sita Botton Suhail Khouh 16/10/25	Conférence HODI	
Marina Jule Augusto Amario Kevin Labigand Pascal Benay 02/10/26	Conférence Xénium	
Hamoun Gueyp 05/03/26	"Myosin 1D promotes de novo induction of zebrafish distal morphogenesis" Ibv	
Julie Harivel 02/04/26	"Identification of novel targets of the TSC2 beta family member NODAL by RNA-seq" Ibv	
Nils Ravel 16/09/26	Brain laterality in Drosophila: from sex to behavior Ibv	
Biocpu meeting Julien Heuer 29/10/26	"Analyzing the inner cells transcriptome up to down markers using scRNA-seq" Zoom	
Herveau BGI Tech Jae li 04/05/26	"Stereo-seq: Spatial transcriptomics solutions" IRCAN	
Biocpu meeting Jeanne Le Guez 3rd PhD Wendie Khouh 3rd PhD 22/05/26 22/05/26	Zoom "Integrated Multiscale modeling of cellular ionic transport" "Multiscale analysis of the pathophysiology of the fragile X syndrome"	